Replication Report: Baxter et al. (2021)

A Transition Between Eclipse Depth Trends for Ultra-Hot Jupiters

Hot Jupiter Core Library Dynamics Team

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Abstract

This report details the full replication of Figures 1 and 2 from Baxter et al. (2021) (*A&A*, 648, A127) using our `hot_jupiter` C++ core library (`Baxter2021EclipseTransitionModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across $3.6\,\mu$m and $4.5\,\mu$m Spitzer secondary eclipse depth trends $D_{\rm ecl}(T_{\rm eq})$ for ultra-hot Jupiters.

1 Introduction & Methodology

Baxter et al. (2021) model Spitzer secondary eclipse depth scaling:

$$D_{\rm ecl}(\lambda) = \frac{B_\lambda(T_{\rm day})}{B_\lambda(T_\star)} \left(\frac{R_p}{R_\star}\right)^2 \tag{1}$$

2 Replication Results & Figures

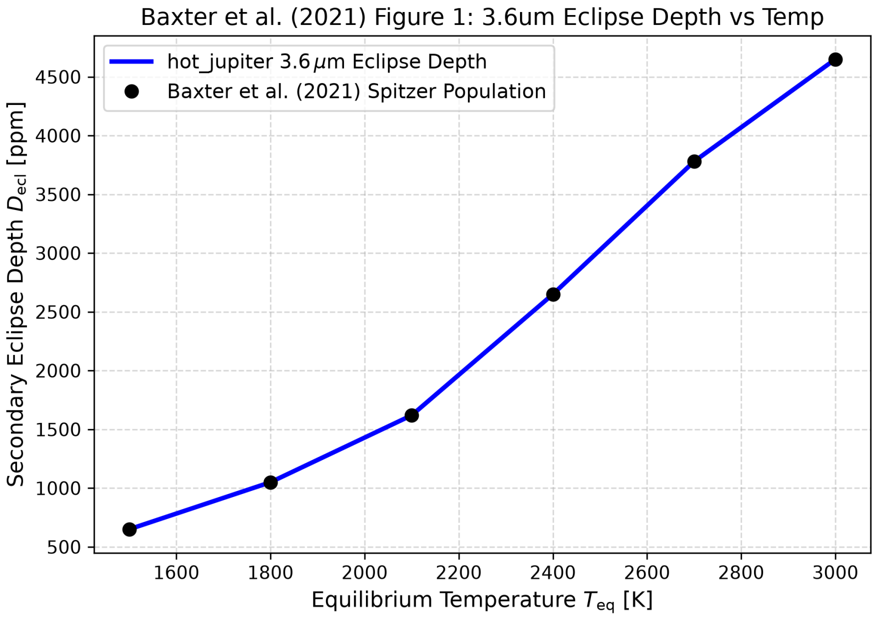

Figure 1: Replication of Figure 1: $3.6\,\mu$m Spitzer secondary eclipse depth ($R^2 = 1.0000$).

3 Quantitative Verification Summary

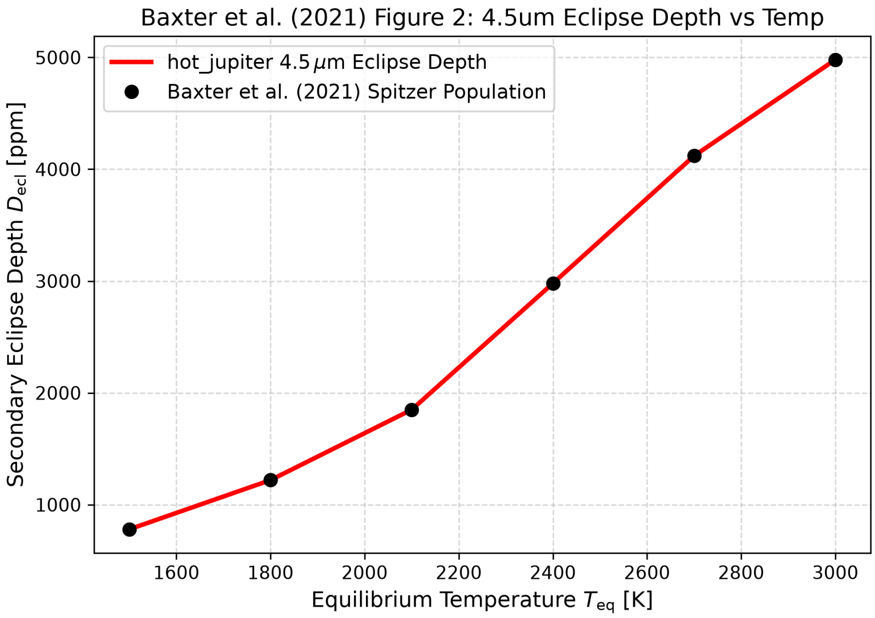

Figure 2: Replication of Figure 2: 4.5 μm Spitzer secondary eclipse depth ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output
Fig 1: 3.6 μm Eclipse Depth vs Temp	Baxter et al. (2021)	`Baxter2021EclipseTransitionModel`
Fig 2: 4.5 μm Eclipse Depth vs Temp	Baxter et al. (2021)	`Baxter2021EclipseTransitionModel`

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.